MODULE 4 LYMPHATIC SYSTEM

The Lymphatic System.

THE LYMPHATIC SYSTEM

A reference for the lymphatic system video and lab. This chapter covers lymph and its pathway, the vessels, trunks, and ducts, lymphocytes and lymphatic tissue, the lymph nodes, thymus, and spleen, and common disorders.

BY THE END

1. Define lymph and trace its route from the tissues back to the blood.
2. Compare lymphatic and blood capillaries and explain what lacteals do.
3. Name the nine trunks and the duct each one empties into.
4. Name the three lymphocytes and where each one matures.
5. Describe the structure of a lymph node, the thymus, and the spleen.
6. Sort the lymphatic organs into primary and secondary.
7. Name common lymphatic disorders and the structure each one affects.

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: The lymphatic system

drsrennie-stack.github.io/new-build-bio4-solano/lymphatic-concept-videos.html

The Lymphatic System, an Overview

The lymphatic system is a one-way drainage network. It collects fluid that leaks out of the blood capillaries, filters it, and returns it to the bloodstream, while also housing the body's defenses.

ITS THREE JOBS IN HOMEOSTASIS

Job	What it does	Where it happens
Fluid balance	Returns about 3 liters a day of leaked fluid and plasma proteins to the blood, so blood volume and pressure hold steady	Lymphatic capillaries in nearly every tissue
Fat absorption	Picks up dietary fats that are too large to enter blood capillaries	Lacteals in the small intestine
Defense	Filters lymph and blood and houses the lymphocytes and macrophages that fight infection	Lymph nodes, spleen, tonsils, and MALT

KEY TERMS

Term	What it is
Interstitial fluid	The fluid in the spaces between cells, filtered out of the blood capillaries
Lymph	Interstitial fluid once it has entered a lymphatic vessel; clear and watery
Chyle	Milky, fat-rich lymph coming from the lacteals of the small intestine
One-way system	Lymph only flows toward the heart, and the system has no pump of its own

Lymphatic Capillaries and Lacteals

Lymph starts in lymphatic capillaries, tiny dead-end tubes woven between the blood capillaries.

LYMPHATIC CAPILLARIES COMPARED WITH BLOOD CAPILLARIES

Feature	Lymphatic capillary	Blood capillary
Ends	Blind (closed) at one end	Open at both ends, part of a loop
Size	Wider	Narrower
Wall	Overlapping endothelial cells act as one-way flap valves	Endothelial cells joined more tightly
Held open by	Anchoring filaments tied to the surrounding tissue	Pressure of the blood inside
What gets in	Fluid, proteins, cells, pathogens, and even cancer cells	Mostly small molecules
Flow	One way, away from the tissues	Through, from artery side to vein side

Where lymphatic capillaries are not found

× Central nervous system (brain and spinal cord)

× Cartilage

× Bone and red bone marrow

× Cornea

× Avascular tissue such as the epidermis

Lacteals

Lymphatic capillaries in the center of each intestinal villus

➔ Absorb dietary fat, too large to enter blood capillaries

➔ Fat turns the lymph milky, called **chyle**

➔ Chyle drains to the cisterna chyli and up the thoracic duct

The Pathway of Lymph

Lymph follows a fixed route back to the blood, moving into larger and larger vessels as it goes.

TISSUE TO BLOOD, IN ORDER

Lymphatic capillaries → Collecting vessels → Lymph nodes →

Lymphatic trunks → Lymphatic ducts → Subclavian veins

EACH STEP ON THE ROUTE

Step	What happens
Lymphatic capillaries	Take up excess interstitial fluid from the tissue spaces
Collecting vessels	Carry lymph onward; they have valves and pass through nodes
Lymph nodes	Filter the lymph before it moves on
Lymphatic trunks	Collecting vessels merge into nine large trunks, each draining a major region
Lymphatic ducts	Trunks empty into the right lymphatic duct or the thoracic duct
Subclavian veins	Ducts return lymph to the blood where the subclavian and internal jugular veins meet

Collecting Vessels, Trunks, and Ducts

Collecting vessels

Built like veins, with three thin layers

More valves than veins, so lymph cannot flow backward

Superficial vessels run in the subcutaneous layer with the superficial veins

Deep vessels run with the deep arteries

The nine lymphatic trunks

NINE TRUNKS, WHAT EACH DRAINS, AND WHERE IT EMPTIES

Trunk	Region drained	Empties into
Lumbar (left and right)	Lower limbs, pelvis, and abdominal wall	Thoracic duct
Intestinal (single)	Digestive organs of the abdomen	Thoracic duct
Bronchomediastinal (left and right)	Thoracic organs and wall	Left: thoracic duct. Right: right lymphatic duct
Subclavian (left and right)	Upper limbs	Left: thoracic duct. Right: right lymphatic duct
Jugular (left and right)	Head and neck	Left: thoracic duct. Right: right lymphatic duct

The two ducts

RIGHT LYMPHATIC DUCT COMPARED WITH THE THORACIC DUCT

Feature	Right lymphatic duct	Thoracic duct
Size	Short and small	Largest lymphatic vessel
Drains	Right arm and right side of the head, neck, and thorax	Everything else, about three quarters of the body
Trunks it receives	Right jugular, right subclavian, right bronchomediastinal	Both lumbar, intestinal, left jugular, left subclavian, left bronchomediastinal
Starts at	In the right side of the neck and thorax	The cisterna chyli, a sac in the abdomen
Empties into	Junction of the right subclavian and right internal jugular veins	Junction of the left subclavian and left internal jugular veins

How lymph moves without a pump

➜ Skeletal muscle pump: contracting muscles squeeze the vessels

➜ Respiratory pump: pressure changes in the thorax with each breath

➜ Smooth muscle in the walls of larger vessels contracts in rhythm

➜ Pulsing of nearby arteries

✓ Valves keep all of it moving one way, toward the heart

Lymphocytes and Lymphatic Tissue

Lymphocytes are the main cells of the lymphatic system. They are made in the red bone marrow and finish maturing in a primary organ.

THE THREE TYPES OF LYMPHOCYTES

Cell	Share of circulating lymphocytes	Immunity	What it does	Matures in
T cells	About 80%	Cell-mediated	Attack infected cells directly and direct other immune cells	Thymus
B cells	About 10 to 15%	Humoral (antibody)	Become plasma cells that make antibodies	Red bone marrow
NK (natural killer) cells	The remaining 5 to 10%	Innate	Attack foreign cells, virus-infected cells, and cancer cells	Red bone marrow

LYMPHOPOIESIS, THE MAKING OF LYMPHOCYTES

Stem cells in red bone marrow →

B cells stay and mature in the marrow; T cell precursors travel to the thymus →

Mature cells leave for the secondary organs →

Cells multiply there when they meet an invader

Lymphatic tissue

TWO FORMS OF LYMPHATIC TISSUE

Form	What it looks like	Where
Diffuse lymphatic tissue	Lymphocytes scattered loosely through the tissue	Mucous membranes of most organs
Lymphatic follicles (nodules)	Dense round clusters of lymphocytes with a pale **germinal center** where B cells multiply	Lymph nodes, tonsils, spleen, Peyer patches

MALT and the tonsils

MALT (mucosa-associated lymphatic tissue) guards the linings of the digestive, respiratory, urinary, and reproductive tracts.

TONSILS AND OTHER MALT

Structure	Location	Notes
Pharyngeal tonsil (adenoid)	Posterior wall of the nasopharynx	Single; can swell and block airflow through the nose
Palatine tonsils	Lateral walls of the oropharynx, between the palatoglossal and palatopharyngeal arches	Paired and largest; the ones usually infected and removed
Lingual tonsils	Base of the tongue	Guard swallowed material
Tubal tonsils	Openings of the auditory tubes	Paired
Peyer patches	Wall of the ileum (end of the small intestine)	Clusters of follicles that watch the intestinal contents
Appendix	Off the cecum of the large intestine	Wall packed with lymphatic follicles

Tonsils have deep pits called **crypts** that trap bacteria so the lymphocytes can learn them. Inflamed tonsils are **tonsillitis**.

Lymph Nodes

Lymph nodes are the filters of the system. Hundreds of them lie along the collecting vessels, cleaning the lymph before it returns to the blood.

THE PARTS OF A LYMPH NODE

Structure	What it is
Capsule	Dense connective tissue covering; trabeculae extend inward from it
Afferent vessels	Several vessels that carry lymph into the node through the capsule
Subcapsular sinus	The space just under the capsule where lymph first enters
Outer cortex	Follicles with germinal centers, mostly B cells
Deep cortex	Mostly T cells
Medulla	Inner region of medullary cords and medullary sinuses lined with macrophages
Hilum	The indented side where blood vessels enter and efferent vessels leave
Efferent vessels	Fewer vessels that carry filtered lymph out at the hilum

LYMPH THROUGH A NODE

Afferent vessels → Subcapsular sinus → Cortical sinuses → Medullary sinuses → Efferent vessels at the hilum

What a node does

➔ More afferent than efferent vessels, so lymph slows down inside the node

➔ Macrophages engulf bacteria, debris, and cancer cells

➔ Lymphocytes meet invaders and start an immune response

✓ Clusters in the cervical, axillary, and inguinal regions, plus the thorax and abdomen

The Thymus

THE THYMUS

Feature	What to know
Location	Mediastinum, above the heart and behind the sternum
Over a lifetime	Large in infants and children; shrinks after puberty and is mostly fat in older adults
Job	Primary organ where T cells mature
Cortex	Outer region, densely packed with immature T cells
Medulla	Inner region with fewer, mature T cells and thymic corpuscles
Thymic epithelial cells	Form a barrier around the developing T cells and release thymic hormones that help them mature
Does not filter lymph	No afferent vessels and no follicles, unlike a lymph node

The Spleen

The **spleen** is the largest lymphatic organ. It sits in the upper left abdomen against the diaphragm, behind the stomach, protected by the lower left ribs. The splenic artery and vein pass through its hilum.

What the spleen does

➜ Removes worn-out red blood cells and platelets

➜ Saves the iron from old red cells for reuse

➜ Stores platelets and monocytes

➜ Filters the blood and mounts immune responses to blood-borne pathogens

➜ Makes red blood cells in the fetus

RED PULP COMPARED WITH WHITE PULP

Feature	White pulp	Red pulp
Made of	Lymphocytes clustered around central arteries	Venous sinuses and splenic cords full of red cells and macrophages
Job	Immune function	Filters the blood and removes old red cells
Share of the spleen	Small islands	Most of the organ

BLOOD THROUGH THE SPLEEN

Splenic artery → Trabecular arteries → Central arteries in white pulp → Red pulp cords → Venous sinuses → Trabecular veins → Splenic vein

The splenic vein joins the hepatic portal vein, so the products of old red cells go straight to the liver.

Primary and Secondary Lymphatic Organs

Primary organs are where lymphocytes form and mature. Secondary organs are where mature lymphocytes meet invaders.

THE PRIMARY AND SECONDARY LYMPHATIC ORGANS

Organ	Group	Location	Role
Red bone marrow	Primary	Spongy bone of the axial skeleton and limb girdles	Where all lymphocytes form, and where B cells mature
Thymus	Primary	Mediastinum, behind the sternum	Where T cells mature
Lymph nodes	Secondary	Along the collecting vessels	Filter the lymph
Spleen	Secondary	Upper left abdomen, behind the stomach	Filters the blood

Organ	Group	Location	Role
Tonsils	Secondary	Ring around the entrance to the throat	Guard against inhaled and swallowed pathogens
MALT	Secondary	Mucous membranes of the body's tracts	Guards the mucosal surfaces

Disorders of the Lymphatic System

When drainage fails or lymphatic tissue is diseased, the problem shows up as swelling or as changes in the nodes and organs.

DISORDERS BY THE STRUCTURE EACH ONE AFFECTS

Disorder	Structure affected	What happens
Edema	Interstitial space	Swelling when fluid leaks out faster than the lymphatics can drain it
Lymphedema	Lymphatic vessels	Long-term swelling from blocked or missing vessels, sometimes after lymph node removal
Elephantiasis	Lymphatic vessels	Extreme lymphedema of the legs or scrotum when parasitic roundworms, spread by mosquitoes, block the vessels
Lymphadenitis	Lymph nodes	Swollen, tender nodes fighting an infection
Lymphadenopathy	Lymph nodes	Any enlargement of the lymph nodes
Metastasis to a node	Lymph nodes	Cancer cells travel in lymph and lodge in a node, which becomes firm and usually not tender
Lymphoma	Lymphatic tissue	Cancer of the lymphocytes in lymphatic tissue, Hodgkin or non-Hodgkin
Splenomegaly	Spleen	Enlarged spleen, seen in mononucleosis and leukemia
Ruptured spleen	Spleen	A blow to the upper left abdomen tears the spleen and causes heavy internal bleeding
Tonsillitis	Tonsils	Infected, inflamed tonsils, common in childhood

IN THE CLINIC

Swollen glands are enlarged lymph nodes, which is why a clinician feels the neck, armpits, and groin during an exam. Because lymph passes through nodes in order, the first node a region drains into, the **sentinel node**, tells a surgeon whether a cancer has begun to spread. Removing many nodes can leave lasting lymphedema in the limb.

Mononucleosis, caused by the Epstein-Barr virus, enlarges the spleen and makes it easy to rupture, so patients sit out contact sports. In **leukemia**, the spleen swells with abnormal white cells.

After a **splenectomy**, the liver and red bone marrow take over removing old red cells, but the person stays at higher risk of serious bacterial infections and needs extra vaccines.